

InsightAI Executive Report

Based on the analysis of 1000 records, the data presents a robust 99.2% health index. We recommend a regression model targeting 'Sales' using the available 4 numeric features. Immediate opportunity exists to clean remaining minor flaws and pilot a baseline model to drive measurable business KPIs.

Dataset Overview

Dataset contains 1,000 rows and 10 columns. 4 numeric fields, 3 categorical fields, and 1 datetime-like fields identified. Identifier columns excluded from analysis: Order_ID, Customer_ID. Health Score: 99.2% (Completeness: 99.2%, Duplicate Penalty: 0.0%).

Health Score

99.2

Key Findings

- The dataset features 1000 rows across 10 columns with 99.2% health score.
- Sales and Profit show the strongest correlation at 0.86.
- Categorical analysis reveals 'Monitor' is the most common segment in Product (241 occurrences).
- Model target candidates include Sales.

Business Opportunities

- With 4 numeric features available, the strong correlation (0.86) between Sales and Profit suggests these variables could be combined or used as interaction terms in modeling.
- Design targeted interventions for the dominant 'Monitor' cohort to maximize impact within the Product segment.
- With Sales ranging from 100 to 5000 (median 1000), automating forecasting could help proactively flag high-value cases for operational decisions.

Risk Factors

- Missing values detected in 4 columns, leading to possible training biases.
- Anomalous values detected: Sales (99 outliers), Profit (65 outliers) (IQR method) could distort predictions.
- Framing as regression requires valid stakeholder alignment on performance threshold.

ML Strategy

Primary path: regression workflow with Linear Regression as leading candidate.